Problem Statement No. 11: Intelligent Shopping Agent

The Challenge

Shoppers today face an overload of product options, scattered reviews, and dynamic pricing across e-commerce platforms. Finding the right product that balances price, quality, and personal preferences is time-consuming and frustrating. Traditional recommendation systems often fail to capture real-time trends, personalized needs, or contextual factors like urgency, budget, or sustainability. The challenge is not just to recommend products but to act as a smart shopping companion that helps buyers make informed and confident decisions.

The Objective

Build an agentic AI shopping assistant that provides intelligent, personalized, and real-time support. The solution should:

1. Fuse Multisource Data — Aggregate product details, reviews, ratings, and price comparisons across multiple e-commerce sites into unified insights.
2. Enable Multimodal Search — Allow users to input voice queries, images (e.g., a photo of a product), or text descriptions to find best-matching products.
3. Create a Predictive & Agentic Layer — Suggest purchase timing (e.g., discounts, seasonal offers), recommend alternatives, and alert users about price drops or fake reviews.
4. Visualize Smart Shopping Pulse — Provide a dashboard summarizing comparisons, sustainability scores, trending products, and personalized shopping lists.

Tech Stack (Mandatory)

IBM Bob + IBM Cloud + IBM Granite